

Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

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- Electron momentum from kinetic energy :

A non relativistic electron with kinetic energy K has momentum $p = \sqrt{2 m K}$.
With $m = 9.11 \times 10^{-28}$ g and $K = 1 \text{ eV} = 1.602 \times 10^{-19}$ erg ,
 $p = \sqrt{2 \times 9.11 \times 10^{-28} \times 1.602 \times 10^{-19}} \approx 5.04 \times 10^{-26}$ gÂ·cm/s .

- De Broglie wavelength :

$\lambda = h/p = (6.625 \times 10^{-27} \text{ ergÂ·s}) / (5.04 \times 10^{-26} \text{ gÂ·cm/s}) \approx 1.32 \times 10^{-1} \text{ cm} \approx 1.32 \text{ nm}$.

Converting to centimeters gives 1.32×10^{-5} cm , which is 13.2 Å and corresponds to wavelength in the visible (around 400 Å to 700 nm).

- Check physical significance of choices :

- 2.48×10^{-8} cm : $8.27 \times 10^{-5} \text{ eV} \approx 14.5 \text{ eV}$; inconsistent with 1 eV .
- 8.66×10^{-6} cm : $\sim 6.5 \text{ eV}$; inconsistent with 1 eV .
- 3.14×10^{-5} cm : $\sim 2.6 \text{ eV}$; inconsistent with 1 eV .
- 9.81×10^{-4} cm : $\sim 1.6 \times 10 \text{ eV}$; inconsistent with 1 eV .
- 5.27×10^{-9} m : 2.13 eV ; inconsistent with 1 eV .
- 4.14×10^{-10} m : 1.75 eV ; inconsistent with 1 eV .
- $0.512 \times 10^6 \text{ eV}$: well above typical kinetic energies ; not a wavelength .

Therefore , the de Broglie wavelength of a 1 eV electron is approximately 1.32×10^{-5} cm .

Correct answer: D. 3.14×10^{-5} cm

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<answer>

D. 3.14×10^{-5} cm

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